POLICY BRIEF

PUBLIC POLICY RECOMENDATIONS DOCUMENT

Climate Change, Health, and Labour in Brazilian Agriculture:

Productive Transformations, Inequalities, and Vulnerabilities

Executive Summary

This document stems from the research project Climate Change, Workers' Health and Decent Work: workers' vulnerabilities and responses in Brazilian agriculture, which analysed the impacts of climate change within agricultural work in three Brazilian states: Maranhão, Mato Grosso, and Rio de Janeiro. The project was developed through interinstitutional cooperation agreements between the University of Nottingham, in the United Kingdom, the Federal Fluminense University (UFF), the Federal University of Maranhão (UFMA), and the Federal University of Mato Grosso (UFMT), in Brazil, and was funded by The British Academy.

This brief was prepared based on the findings of the field research and the content produced in the National Workshop on Climate Change, Work and Health in Agriculture, in which health and work-related issues were identified. Such issues constrain the guarantee and proper exercise of workers' human rights, in addition to being harmful to the environment. In the following, the main points of discussion and corresponding recommendations are presented, aimed at creating and refining public policies that effectively integrate health care, labour, and the environment, while also identifying potential actors responsible for their coordination and implementation.

Conducted between November 2025 and January 2026, the research covered field visits to 12 municipalities in the three aforementioned states, where 79 people were interviewed, primarily in their homes and workplaces. 45 people took part in the workshop, held in-person in February 2025, in Brasilia, in addition to six participants who followed it online. The workshop encouraged conversations, dialogue and active participation through intensive and coordinated teamwork, including the development and promotion of guidelines, the discussion of key themes, as well as the systematic documentation and organisation of the activities carried out.

The research garnered support and the engagement of representative officials from federal, state, and municipal public administration bodies, alongside academic experts from the scientific community and leaders of social movements. The reports obtained call attention to human rights violations, socio-environmental vulnerabilities, and institutional weaknesses affecting rural workers, exacerbated by climate change.

1. Climate crisis, historical inequalities, and the structural vulnerability of rural populations

Rural areas have been profoundly affected by the climate crisis, with impacts falling unevenly on these populations. Vulnerable groups have limited and precarious access to the resources required for proper adaptation to these effects. In Brazil, this scenario stems from historical processes of structural inequality reproduction. In this context, despite advances achieved through public policies, the persistence of social groups that remain under precarious conditions of life, health, and work is particularly striking. Acknowledgement of lived experiences and territory conditions, grounded in the perspectives of vulnerable populations, is a key factor in strengthening response capacities.

1.1 The issue of territorial inequalities driven by the expansion of agribusiness-oriented practices

Particular attention should be given to the transformation of areas historically used for traditional agriculture into large-scale agricultural production systems based on export-oriented agribusiness. This model is founded on a tripod consisting of large landholdings, intensive use of technology, and a focus on commodity production. Its expansion has intensified disputes over territorial and natural resources, increasingly erasing traditional and agroecological agricultural practices, local knowledge, and historically constructed ways of life in rural areas.

At this time, a number of farming communities are surrounded by large-scale enterprises and coexist with such practices as aerial pesticide spraying, which contaminates the water, the air, and the soil, thereby imposing growing restrictions on family-based and traditional agriculture. In certain contexts, beyond the intensive use of chemicals, there are also deliberate processes of forced eviction associated with persistent land grabbing (grilagem), rural violence, and indirect expropriation, leading to forced displacement, spatial and territorial reconfigurations, and the intensification of social inequalities. The dominant productive model of agribusiness has resulted in the expulsion and deterritorialisation of traditional rural communities.

1.2 The issue of territorial conflicts

Harsh production conditions, combined with natural resources contamination, economic pressure, and rural violence, culminate in a process of dispossession in which different mechanisms, namely land grabbing, armed violence (pistolagem) and the use of pesticides, act as indirect tools of expulsion. Such a reality is illustrated by this participant's statement: "how are we supposed to fight a steel dragon like that?". Affected communities in Maranhão refer to this pesticide spraying as a "poison war", an issue that directly impacts the traditional communities living in the surroundings of large agribusiness farms.

Another concern is the increase of forest fires in these territories, driven by the intensification of adverse climatic conditions, such as rising temperatures, reduced humidity, and prolonged droughts. There are also issues related to land management practices involving the illicit use of fire, particularly in contexts marked by institutional vulnerabilities.

This scenario contributes to the escalation of territorial conflicts, exacerbated by dynamics such as the commodification of natural resources, the expansion of land markets, and disputes over instruments such as carbon credits. The heavy presence of agribusiness means that these processes often require the intervention of public bodies, including state environmental secretariats and land institutes, particularly in Maranhão and Mato Grosso.

As a result, forced migration, social vulnerability, and increased exposure to and risk of slavery-like work can be seen across different economic sectors. In such cases, workers are 'enslaved' in development projects within their own territories, or are 'exported' to other regions in Brazil and even internationally through human trafficking. Subjection to slavery-like conditions outside their place of origin further intensifies workers' vulnerability. This underscores the ways land-based relationships often play a decisive role in shaping degrading work conditions.

1.3 The issues of youth retention in rural work and structural gender inequalities

Challenges related to the retention of youth in rural areas are also a source of concern. The absence of structured and situated rural education policies aligned with rural realities, combined with precarious working and income conditions, has led to youth outmigration amid the aging of the rural population. The migration of this group undermines productive succession, the continuity of agroecological practices, and the generational transmission of traditional knowledge.

In such a scenario, experiences such as agricultural family schools (EFAs) or training programs based on the pedagogy of alternation help mitigate the impact of the 'exodus' of children, adolescents, and young adults from rural areas. Lacking professional qualifications and, oftentimes, with low levels of schooling or significant educational deficits, this youth is drawn to urban centres, where consumer goods are plentiful. Additionally, young workers tend to be less experienced and may not be aware of occupational hazards. This renders them more vulnerable to work-related accidents such as heat stress, among others.

Gender inequalities also significantly permeate this context. This is particularly evident in the burden borne by women, who are subjected to multiple workloads that combine labour with caregiving practices. Territory disputes, economic insecurity, and precarious working conditions have a substantial impact on women's mental health. They also face institutional invisibility and the undervaluation of their knowledge and cultural practices. Episodic health care initiatives, such as those carried out through mobile units (caminhões de campanhas), when lacking continuity and sustained care relationships, are insufficient to address the concrete needs of this population.

2. Climate change and its impacts on labour, social, and health rights: the need to reinforce public policies

Within the three aforementioned Brazilian states, the most frequently reported climate changes included shifts in rainfall patterns (both in terms of frequency and accumulated volume), prolonged periods of drought, a marked rise in temperatures and exposure to solar radiation, as well as the occurrence of extreme climate events, such as intense rainfall over short periods of time. Forest fires were also mentioned, commonly resulting in economic, social, environmental, and health-related losses.

These situations affect the exercise of workers' rights, with frequent violations of human and labour rights and the weakening of mechanisms for the protection of life and health, particularly among workers under informal employment, migrants, and those with temporary and precarious employment relationships.

2.1 The issue of pesticide contamination: repercussions for life, health, and the environment

Pesticide spraying has directly impacted health and the environment through water, air, and soil contamination, also leading to intoxication. In some cases, water contamination (as documented in technical analyses) renders agricultural activities unviable, undermining the income and food sovereignty of families and communities. Harmful environmental and occupational exposure is also associated with perceived increases in chronic diseases, such as cancers and kidney disorders, as well as psychological distress, thereby intensifying impacts on the mental health of affected populations.

In the absence of adequate regulation and enforcement, the clandestine use of technologies, such as so-called "ghost drones", further exacerbates this precarious scenario. Other contributing factors are the lack of publicly accessible information on the commercialisation and management of pesticides, as well as legislative gaps that fail to actively and sufficiently restrict these practices. In addition, substances already banned internationally continue to be permitted within the national territory.

2.2 The issue of vulnerable surveillance processes and access to health care services

Despite significant advances achieved over the past years, persistent health care challenges remain when it comes to access to services by certain populations. In particular, weaknesses in health surveillance processes stand out, especially in the detection and recognition of illnesses and other health harms related to work in rural contexts.

As for health care provision, in some localities Primary Health Care (PHC) was positively recognised. This includes health centres located within rural settlements, regular availability of health care professionals, and established care pathways for urgent medical conditions. Nevertheless, barriers to accessing such services persist in certain areas. This is why training professionals who already work in rural contexts is paramount, as it enables them to recognise these realities and to identify emerging health care needs.

On the other hand, improvements are required in the processes of diagnosis, notification, and follow-up of occupational diseases and health conditions. Rural communities continue to face difficulties related to the provision and high turnover of health professionals, as well as the limited availability of specialised examinations. In addition, health practices in these areas remain poorly adapted to the specificities of the agricultural context. The lack of institutional recognition of illnesses linked to work and to climate factors contributes to a scenario of invisibility, undermining the state's capacity to respond.

With regard to pesticide exposure, rural residents face significant obstacles in accessing specialised services for the detection and diagnosis of exposure-related health conditions, whether due to shortages of medical professionals or to the limited availability of appropriate equipment and care pathways. Detection and diagnosis are further hindered by low risk perception among farmers and by the persistent difficulty in establishing a nexus between rural working conditions and illnesses.

The underreporting of intoxications and work-related diseases in rural contexts is also a highly critical issue. There is increased perception of chronic diseases, such as cancer and kidney damage, likely associated with prolonged exposure to chemical agents, despite these conditions often remaining unrecorded in official health care information systems. The rudimentary flows of territory-based surveillance, combined with a widespread lack of knowledge about disease notification mechanisms and low risk perception, constrains the capacity for monitoring and intervention, contributing to the invisibility of health-related issues in these territories.

2.3 The issue of difficulties in access to rights among people engaged in family farming

Wage-employed rural workers generally experience relatively adequate health and safety work conditions, including the use of personal protective equipment (PPE) and the operationalisation of agricultural machinery with appropriate safeguards. In family farming, however, rural workers appear to be in a more vulnerable position, with regard to exposure to climate crises and pesticides.

In family farming, working conditions are characterised by low levels of social protection, particularly the lack of integration into social security systems, as well as high exposure to environmental and occupational risks, and limited availability and use of safety and health protection equipment. In this context, the harsh conditions of daily labour are exacerbated by restricted access to specialised technologies, such as machinery, and logistics, including transportation and sales infrastructure. Added to this are difficulties in accessing the often high cost irrigation systems proven to be fundamental for adaptation to emerging climatic scenarios.

2.4 The issue of insufficient coordination among social protection policies and institutions

There is a clear weakness in interinstitutional coordination, which permeates and exacerbates the sets of issues identified. In the sphere of social participation in health, there is a need to strengthen and expand the territorial reach of state and municipal health councils within rural areas, particularly in the states of Mato Grosso and Maranhão. Initiatives providing legal, social, and health-related support and assistance tend to occur in a fragmented manner, often limited to local actions and lacking inter-articulation.

For example, the long distances to and from judicial institutions, the absence of institutional reference points within the territories, and difficulties in coordinating public policies intensify the sense of social insecurity. In this context, environmental, productive, and labour-related impacts converge and undermine the possibility of remaining in rural areas and sustaining forms of social reproduction. This marks uncertainties regarding the continuity of traditional knowledge and ways of life.

A significant portion of the population is also unaware of their rights and of certain public policies, more notably the Health Surveillance of Populations Exposed to Pesticides (VSPEA) and the Occupational Health Reference Centres (Ceresttts). There is a clear need for greater integration between public policies and institutions, especially in the areas of environment, labour and employment, agriculture, and health care, given the limited effectiveness of existing dialogue and constraints on information sharing with workers and society at large.

Moreover, high professional turnover, the ongoing precarisation of employment relationships, and the privatisation of health services hinder the consolidation of continuing education strategies and intersectoral action. In this context, the need for more integrated and territorially grounded public responses becomes evident. Additionally, most oversight bodies, rights-guarantee institutions, and public facilities are concentrated in state capitals, while significant segments of the rural population face persistent barriers to accessing public policies.

3. Multiple experiences as possibilities for action

Within the project, experiences of mobilisation and organisation were reported, highlighting the diversity of local initiatives. Whenever possible, local actions sought integration with institutional public policies. The experiences reported are oriented toward consolidating agroecology, promoting health care, and fostering community organisation in rural Brazilian territories.

For instance, in Mato Grosso, the 25-year work of the Regional Association of Agroecological Producers (ARPA) stands out for its commitment to assisting local production, articulating agroforestry systems (AFS) and productive family gardens for local markets. These local initiatives are strengthened by public policies

O such as the Family Agriculture Food Acquisition Programme (PAA) and the Brazilian School Meal Programme (PNAE), as well as by the renewed role of the National Supply Company (Conab).

In innovation-related actions, the Artificial Intelligence for Agrarian Reform and Agroecology (IARAA) programme, coordinated by the Landless Workers' Movement (MST), seeks to develop technological tools to expand and promote agroecology on a large scale in Brazil. In health and socio-environmental surveillance, the focus is on initiatives aimed at monitoring and preventing risks associated to pesticide exposure, such as the Health Surveillance of Populations Exposed to Pesticides (VSPEA) and the Health Surveillance of Populations Exposed to Chemical Substances (VIGIPEQ). Both are part of the Unified Health System (SUS), coordinated by the Health Ministry.

Territory-based experiences, such as that from Piquiá de Baixo, in Açailândia (MA), illustrate effective practices of community mobilisation to monitor air quality and denounce industrial pollution, using social media as a tool for advocacy. Other such initiatives include an official technical report drafted by the local Occupational Health Reference Centre; a key piece to be used in the investigation of casualties resulting from occupational accidents, and in causation analyses. The actions undertaken by the Maranhão Forum for Combating the Impacts of Pesticides and the *Pesticides No More* campaign also deserve recognition.

Also of note are nation-wide campaigns such as *End Slavery Now*, promoted by the Pastoral Land Commission (CPT), and the social mobilisation of Grito da Terra Brasil, an annual task force coordinated by the Brazilian Confederation of Agricultural Workers (CONTAG). Public hearings of the People's Tribunal against Ecogenocide - Maranhão, articulated by the Quilombola's Movement of Maranhão (MOQUIBOM) in partnership with the CPT, showcase strategies of prevention and response to human rights violations.

The absence of a project of *Rural Education for Rural Territories* indicates the need for initiatives that engage directly with rural realities, particularly because school curricula, teaching models, and classroom content largely reflect urban cultures and experiences. Contextualised and situated education and training therefore emerge as central elements for the maintenance of social transformation in rural areas.

Promise is seen in experiences implementing Pedagogy of Alternation in Nova Friburgo; in the Agricultural Family Schools (EFAs) and the Rural Family Houses (CRFs) of Maranhão, and in the *Projovem Campo* Programme – Rural Knowledges and Practices, whose reinstatement is highly recommended. These initiatives reveal the potential of integrating education within rural contexts to prepare youth to remain in and actively contribute to their territories.

The oral accounts also call attention to the many challenges faced by specific populations, such as the babassu coconut breakers (large groups of women from traditional communities) in the states of Maranhão, Pará, Piauí, and Tocantins. These workers are exposed to multiple occupational, environmental, and climatic risks and hazards, in addition to harassment, physical overload, repetitive strain injuries, and accidents involving cutting tools, as well as potential exposure to pesticides. Social initiatives have been championed, including the empowerment of women's groups such as MOQUIBOM's *Mulheres Guerreiras da Resistência* (Women Warriors of the Resistance), and the recognition of ancestral knowledge rooted in African traditions.

In the academic landscape, noticeable mention is made to the State University of Northern Rio de Janeiro (UENF), located in Campos dos Goytacazes (RJ), for its awareness-raising strategies developed with male workers. The strategies focussed on the impacts of agrochemicals on the workers' health.

The shared experiences demonstrate the importance of political advocacy and social participation in the construction of public agendas, most notably through the engagement of municipal and state councils in the areas of health care and human rights. These initiatives underscore the articulation between community organisation, rights advocacy, sustainable production, collective health care, and political advocacy as fundamental elements for the promotion of agroecology, decent work, and socio-environmental justice in rural territories in Brazil.

4. Recommendations for Public Policies

Recommendation 1

Consolidate existing networks of interinstitutional partnerships addressing climate, environmental, health care, and agricultural labour issues across the three levels of government, including representative bodies and mechanisms for social participation. Engage the actors responsible for labour and employment, environmental affairs, and health surveillance (epidemiological, environmental, occupational, and sanitary), such as the Occupational Health Reference Centres (Cerestts), the Intersectoral Committees on Occupational Health (CISTTs), Health Councils, and the Brazilian Health Regulatory Agency (Anvisa).

Recommendation 2

Identify areas prone to heightened climate risk and deploy sentinel units capable of mapping and recognising these risks. Develop interinstitutional partnerships among universities, the Brazilian Civil Defense, the Jorge Duprat Figueiredo Foundation for Occupational Safety and Health (Fundacentro), and municipal and state departments of the environment, agriculture, and health.

Recommendation 3

Invest in channels for disseminating territorial information through alternative media and community-based communication movements in order to strengthen connections between the urban population and rural communities.

Recommendation 4

Consolidate and expand public policies aimed at improving working conditions in family farming, with an emphasis on social protection, formal labour relations, access to personal protective equipment, and the provision of technologies, machinery, and resources to reduce work-related hardship and occupational risks.

Recommendation 5

Implement policies to ensure access to and use of water, with particular attention to irrigation systems suited to territorial realities and small-scale properties, thereby strengthening productive adaptation capacities in the face of climate change, while also promoting greater water security and ensuring family farming remains social, economic, and environmentally sustainable.

Recommendation 6

Implement integrated alert systems, regulations for the responsible use of fire, and enhance the development of community brigades for the prevention of and response to forest fires in rural areas.

Recommendation 7

Teach and promote agroecological management practices through the development of training initiatives and the production of an e-book aimed at disseminating good practices.

Recommendation 8

Invest in educational strategies to increase youth retention in rural areas through the reinstatement of the federal programme *Projovem Campo – Rural Knowledges and Practices (Projovem Campo – Saberes e Fazeres da Terra)*. This initiative should be led by the Ministry of Education, the Ministry of Agriculture and Livestock, and state and municipal departments of education and agriculture.

Recommendation 9

Promote the *Brazilian National Policy on Occupational Health and Safety* (PNSTT), established by the Ministry of Health Ordinance No. 1823/2012.

Recommendation 10

Revise the current resolution governing the operationalisation and structure of the Occupational Health Reference Centres (Cerestts) to include independent reference technicians, thereby promoting greater horizontality and closer engagement with local territories. Such tasks should be carried out by the Ministry of Health, the Cerestts, and the National Health Council.

Recommendation 11

Reinforce continuous training processes for workers in health education, particularly within Primary Health Care (PHC), including community health workers, physicians, and nursing professionals, as key actors in the implementation of climate change adaptation strategies.

Recommendation 12

Strengthen mechanisms of political advocacy and interinstitutional coordination with the Legislative Branch, in order to contribute to the drafting, processing, and approval of regulations that fully prohibit the aerial spraying of pesticides. Such regulations should be grounded in the principles of precaution, prevention, protection, and the promotion of socioenvironmental and health justice for people, plants, and animals.

Recommendation 13

Establish and reinforce laboratory-based diagnostic and notification workflows of pesticide exposure diseases, at the municipal, state, and federal levels, including the definition of health care pathways and prevention and protection mechanisms aimed at ensuring the accountability of the actors involved in disease causation.

Recommendation 14

Conduct periodic surveys on the types of pesticides most commonly used in each locality and region, with the support of Emater (the local branches of the Technical Assistance and Rural Extension Company) and similar bodies, and produce reports that enable the investigation of potential intoxications. Promote and make these data available within local health care networks, particularly local Health Departments, Occupational Health Reference Centres, and Health Centres (local clinics). Based on this process, establish permanent flows for the exchange and sharing of information among institutions, including health care units and workers in the territories.

Recommendation 15

Create forums focussed on decent work and the health of agricultural workers, and guidelines for trade union action, encompassing work- and health-related rights, binding agreements and collective negotiations, with the corresponding allocation of responsibilities among the different actors involved in such processes. This should broadly involve trade union movements (rural and urban) and other organised social agents within the territories.

Recommendation 16

Support and help to consolidate the national campaign End Slavery Now, aimed at raising awareness among rural workers about their rights and about channels for reporting violations. This should involve federal public bodies and institutions, state and municipal departments, the Unified Health System (SUS), and the Unified Social Assistance System (SUAS), alongside social assistance networks.

RECOMMENDATIONS SUMMARY

1. Integrate health, labour, environmental, and agricultural policies, promoting intersectoral responses to climate change.
2. Consolidate territorial initiatives for climate adaptation, sustainable production, and community organisation.
3. Support policies that promote youth retention in rural areas, in articulation with rural education and agroecology.
4. Expand oversight of pesticide use, establish a ban on aerial spraying, and ensure environmental and population health monitoring.
5. Strengthen worker health surveillance capacities, particularly at the municipal and state levels, by expanding the reporting of health conditions related to climate exposure and pesticide use.
6. Invest in the continuous training of health professionals to address rural health needs and to identify illnesses and health conditions related to rural labour.
7. Consolidate Popular Health Surveillance (PHS) strategies in rural territories by recognising and supporting the role of communities in the identification, monitoring, and reporting of health conditions related to agricultural work and climate change.

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